University of Central Punjab
Faculty of Information Technology

Assignment 1

Course Title: Artificial Intelligence

Name	Registration#

Submission Requirements

1. **Format:** The assignment must be **handwritten** and submitted as a single document (PDF or Word recommended) in softcopy.
2. **Integrity:** This is an **individual assignment**. Any evidence of plagiarism or sharing of assignment will result in a zero grade.
3. **A quiz will be conducted and will carry the weightage for assignment marks.**
4. **Due Date for Assignment is October 26th, 2026, 6PM.**
5. **NO LATE SUBMISSION.**
6. **NO ASSIGNMENT WILL BE ACCEPTED VIA EMAIL. PLEASE BE CAREFUL ABOUT TIMELY SUBMISSION.**

CLO#	Course Learning Outcome (CLO)	Taxonomy Level	Mapping to PLO
CLO 1	To understand and explain the solid fundamental principles and concepts of artificial intelligence, including machine learning and neural networks.	C2	PLO1

Total Marks : 10 x 8= 80

Question 1:

Prepare a PEAS description for an AI-powered university campus delivery robot. Include at least three suitable items under each heading: Performance Measure, Environment, Actuators, and Sensors.

Question 2:

For each environment below, classify it using the indicated property and justify your answer in one sentence:

i. Chess: fully observable or partially observable?

ii. Taxi driving: deterministic or stochastic?

iii. Crossword puzzle: static or dynamic?

iv. Chess: discrete or continuous?

v. A video game with several competing players: single-agent or multi-agent?

vi. A medical diagnosis system that cannot directly observe every condition inside a patient: fully observable or partially observable?

Question 3:

Compare **BFS and DFS** in terms of:

- Completeness
- Optimality
- Time Complexity
- Space Complexity

Question 4: Explain **Depth-Limited Search** and its limitations. How does **Iterative Deepening Search (IDS)** overcome these limitations?

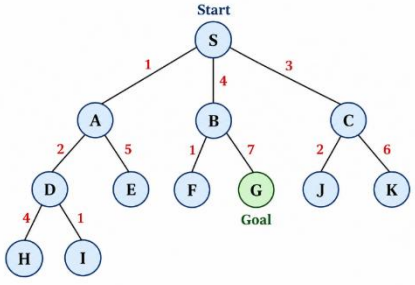
Question 5: Uninformed Searching

Consider the following graph where 'S' is the start node and 'G' is the goal node. The numbers on the edges represent the path costs.

1. Trace the Uniform Cost Search (UCS) algorithm to find the optimal path from S to G.
2. Show the frontier (priority queue) and the explored set at each step.
3. What is the final optimal path and its total cost?

For each of the following search strategies:

1. **Breadth-First Search (BFS)**
2. **Uniform Cost Search (UCS)**



You are required to:

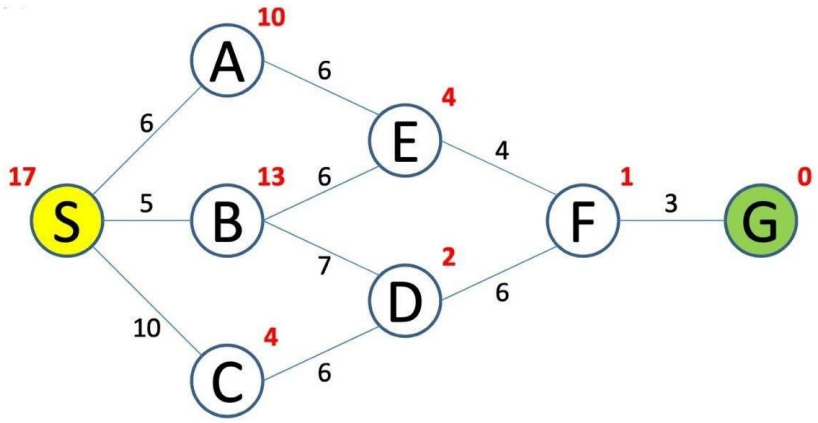
(a) Write the sequence of nodes visited (traversal order)

(b) Determine the final solution path from S to G

(c) Calculate the total path cost (for UCS)

Question 6:

Perform BFS on the graph shown. Explicitly show queue at each step along with a set of explored nodes so far.



Question 7:

Explain the concept of Iterative Deepening Search (IDS).

How does it combine the advantages of Depth-First Search (DFS) and Breadth-First Search (BFS)?

Is IDS optimal? Under what condition?

Question 8:

Algorithm	**Complete?**	**Optimal?**	**Time Complexity**	**Space Complexity**
DFS				
BFS				
UCS				
IDDFS				

BEST OF LUCK 😊